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July 23, 2026

Dear Editors of Knowledge and Information Systems,

Please consider my manuscript, entitled "Retrieval-Assisted Instantiation of Natural-Language Optimization Problems," for publication in *Knowledge and Information Systems*.

Engineering and decision-support settings often begin with optimization problems stated in natural language and only later translated into formal models. That translation remains difficult: recovering the right problem structure and mapping textual quantities onto the correct parameters are both nontrivial, and end-to-end automation readily conflates them. This manuscript studies a focused intermediate problem rather than claiming full natural-language-to-solver automation. Given a natural-language description, the task is to retrieve a compatible optimization schema from a fixed catalog and then instantiate that schema's scalar parameters from numeric evidence in the text.

The approach is transparent and reproducible. Schema retrieval uses deterministic lexical methods; scalar parameters are then filled by schema-conditioned, rule-based grounding that combines numeric extraction, coarse type inference, and compatibility-based slot assignment. On the NLP4LP benchmark, schema retrieval is already strong in the studied setting, while an oracle with perfect retrieval yields only a modest gain in end-to-end instantiation readiness. The main scientific finding is that semantic number-to-slot grounding—not schema selection—is the dominant remaining bottleneck once a plausible schema is available.

The manuscript supports this diagnosis with robustness checks, statistical significance testing, an error analysis, and restricted structural and solver-backed validation studies that are carefully scoped and reported transparently. Implementation and reproducibility artifacts are publicly available at `https://github.com/SoroushVahidi/combinatorial-opt-agent`.

I believe the work fits *Knowledge and Information Systems* because it treats natural-language optimization understanding as a knowledge-processing problem: information retrieval over structured schemas, typed representation of optimization parameters, and semantic grounding of numeric evidence within an intelligent information system. The contribution is a controlled, inspectable account of what deterministic retrieval-assisted instantiation can and cannot yet achieve—useful decision support without overclaiming solver-ready generation, benchmark-wide solver readiness, broad deployment, or superiority to large-language-model systems.

This manuscript is original and is not under consideration elsewhere.

Thank you for considering this manuscript for publication in *Knowledge and Information Systems*.

Sincerely,

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